ME60012- CFD- Assignment 2, Due date 26.02.25

Consider steady heat conduction through a plate of steel material (conductivity 30 W/m-K).

I. Two sides are insulated, the right edge is exposed to airflow at constant temperature with given convective heat transfer coefficient and the bottom edge is heated to a temperature of $T=200x(1-x)$°C in Figure (I). Plot the temperature distribution in the plate. Solve using Gauss-Siedel with SOR. Show grid independence by comparing temperature profile along (0,0)-(1,1) diagonal for at least three different mesh. Determine the optimum SOR factor for each mesh.

II. Now, consider that the geometry is modified as per Figure (II). Plot the temperature distribution in this domain. You may consider the finest Δx and Δy for which grid independence was obtained in part-1. What are the respective temperatures at (0.15,0.15) and (0.9,0.9) ? (The hashed edges are insulated.)

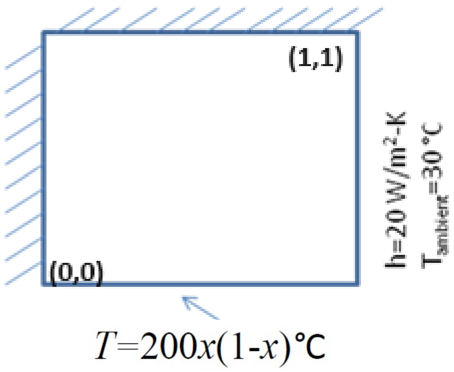

Fig I

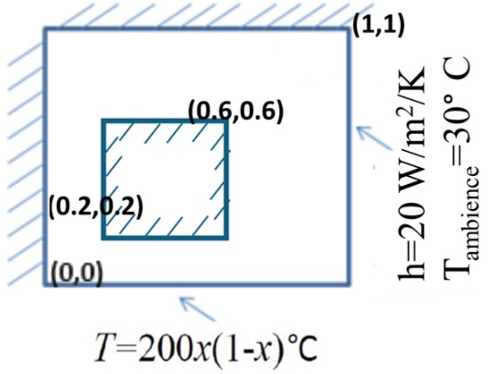

Fig II